Early Prediction of Kidney Dysfunction in Diabetes Patients Based on Fasting Blood Sugar and Creatinine Status

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***Abstract*—This study investigates the early prediction of kidney dysfunction in diabetic patients by analyzing fasting blood sugar and creatinine levels with the use of machine learning models. Two primary machine learning algorithms, namely, k-nearest neighbors and random forest, were developed for the early prediction of kidney dysfunction in diabetic patients based on threshold-derived labels using fasting blood sugar and creatinine levels. Additionally, logistic regression and support vector machine were utilized for comparative evaluation. Since the dataset did not include an entry stating the diagnosis of kidney dysfunction, the target variable was generated using threshold-based rules on the fasting blood sugar and creatinine values. As a result, the models are trained on the labeled data and then applied to predict the derived labels, automating an early-detection screening protocol. A dataset consisting of 499 samples with 13 features was cleaned, integrated, and transformed in three stages. The resulting cleaned two data frame sizes, 310x13 and 496x13, were analyzed for gender-specific distribution of creatinine and fasting blood sugar levels. The analysis revealed that a larger proportion of female patients had creatinine levels exceeding the high-threshold value compared to male patients. Furthermore, the bias in gender distribution in the first data frame affected model performance; alternatively, in the second data frame, the original distribution between males and females was maintained, leading to more consistent results. The analysis proved that females are more likely to develop early-stage kidney dysfunction than males, with a higher rate of kidney dysfunction in diabetic women. On the first dataset, RF achieved the highest performance with a scoring accuracy of 98.39% and a perfect ROC-AUC of 1.0, while KNN scored only 85.48% accuracy with a ROC-AUC of 0.86. On the second dataset, RF accuracy remained unchanged with a value of 98.39% and ROC-AUC = 1.0; on the other hand, the KNN accuracy improved to 93.0% with ROC-AUC = 0.93 due to balanced gender distribution. LR and SVM also achieved competitive results. On the first dataset, LR obtained 95.16% accuracy with a ROC-AUC value of 0.977, while SVM obtained an accuracy of 91.93% and ROC-AUC = 0.963. On the second dataset, LR again achieved 95.16% and ROC-AUC = 0.977, while SVM recorded 91.93% accuracy and ROC-AUC = 0.963.**

***Keywords—Diabetes, KNN, RF,* Kidney Dysfunction**

I. INTRODUCTION

Diabetes is considered a leading cause of kidney dysfunction, affecting the human body silently until irreversible damage occurs [1]. Early prediction of kidney impairment in diabetic patients is urgent to provide suitable health care and minimize the side effects as much as possible. Fasting blood sugar and creatinine levels are biomarkers that indicate the assessment of renal health. This study aims to use the correlation between fasting blood sugar and creatinine status to develop a predictive model able to detect early kidney dysfunction in diabetic patients.

Glucose is considered a premium fuel for the human body, basically obtained from carbohydrates. Glucose is handled by the digestive system, which breaks down glucose within the carbohydrates, creating the glucose [1]. Regulation depends on two components: the insulin hormone and the insulin receptor. The insulin hormone is a hormone produced by the pancreas, and the insulin receptor is located on the surface of muscle cells [2].

Hormone and receptor are mapped as lock and key, where the hormone is the key, and the receptor is the lock. When the insulin binds with the receptor, the glucose carried by the red blood cells can enter the cells, so the power cycle can start. But in some cases, the glucose is stacked in the bloodstream, causing a high level of sugar in the blood, resulting in serious issues through time, ending with diabetes. Diabetes is defined as a collection of metabolic diseases characterized by hyperglycemia due to defects in insulin secretion, insulin working, or both. In some cases, it can lead to failure in the eyes, kidneys, nerves, heart, and blood vessels. Diabetes is classified into two types based on the factors causing its development: type one, which is β-cell destruction leading to absolute insulin deficiency, and type two, which can vary from predominantly insulin resistance with relative insulin deficiency to predominantly an insulin secretory defect with insulin resistance [3]. In type one diabetes, the pancreas does not provide sufficient amounts of the insulin hormone or does not produce it at all. As a result, the glucose remains in the bloodstream, increasing its concentration. Patients can develop type one diabetes at an early stage of life, more frequently in childhood, but it can be developed at any age later on [4]. On the other hand, type two diabetes is defined as insulin resistance. Receptors become less sensitive to the produced insulin, causing the glucose to remain in the bloodstream, increasing its concentration over time. As a response to the high concentration of glucose in the bloodstream, the pancreas produces more insulin, leading over

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time to a reduction in the pancreas's ability to produce insulin as a result of the continuous production. The pancreas is tasked with the production of the insulin hormone whenever the blood contains glucose as part of the blood circulation, then the ratio and concentration of sugar in the blood are balanced [3]. The human body can inherit diabetes as an outcome of bad habits related to the lifestyle itself, such as obesity, poor eating habits, lack of physical activity, smoking, and alcohol consumption. Prevention of such a disease: the patient must monitor their weight, keep it in the normal range, eat healthy, be physically active, and quit smoking [5] [6].

This section provides an overview of diabetes. The rest of the paper is structured as follows: Section II reviews previous literature on diabetes and the implementation of machine learning models in diabetes prediction. Section III enlightens the research methodology, which involves four key stages: data collection, data preprocessing, exploratory data analysis, and data modeling. Section IV displays and investigates the results, including visualizations of male and female diabetic patients, as well as the performance evaluation of the applied machine learning models. Finally, Section V summarizes the study's key findings

II. LITERATURE REVIEW

A significant reduction in the risk factor and severity of diabetes is possible through a precise early prediction model [7]. Machine learning techniques proved their efficiency in forming powerful and accurate prediction models serving as assisting tools for healthcare professionals in detecting diseases specifically [5].

The study in [7] reflects the crucial and rising challenges of accurately predicting diabetes, which is considered a chronic and increasingly prevalent disease worldwide. To overcome such a challenge, a solid machine learning algorithm is introduced by the authors, which uses the strengths of multiple classifiers through an ensemble approach of multiple machine learning classifiers, such as the k-Nearest Neighbor (k-NN), Decision Trees, Random Forest, Naive Bayes, AdaBoost, XGBoost, and Multilayer Perceptron (MLP). Also, suggesting a missing value handling strategy starting with mean imputation, data standardization to ensure zero mean, ending with unit variance, and outlier rejection criteria using the interquartile range (IQR) method.

Authors in [5] investigated the ability of machine learning algorithms to predict raised blood sugar levels, one of the key indicators of prediabetes and diabetes disease infection. Marking the importance of early detection to prevent severe health complications such as cardiovascular diseases and kidney failure. Also, the results proved promising abilities in the early prediction of diabetes by implementing advanced machine learning algorithms. Such as Random Forest, Decision Tree, AdaBoost, XGBoost, Bagging Decision Trees, and Multi-Layer Perceptron, the models achieved high predictive accuracy as follows: (98.4%), (97.4%), (96.4%), (96.3%), (95.2%), and (94.8%).

As the burden of chronic diseases increases, especially diabetes, a demand for innovative solutions to increase public awareness, especially in underprivileged communities, is required. Where the authors in [8] enlightened the ability of exploratory data analysis and data visualization in simplifying complex datasets and making insights accessible to non-expert audiences and uneducated persons by replacing cognitive challenges with clarity and helping the audience to find and understand patterns, anomalies, and trends. The authors used distinguishable color codes like red, green, and yellow, aiming to improve the community's ability to understand test results without being highly educated.

High levels of fasting blood sugar trigger multiple factors that leads to kidney failure. Authors in [9] pointed to the continued increase in hyperglycemia as the main cause of glomerular hyperfiltration and oxidative stress. Resulting in both structural and functional damage to the kidney. The glomerular hyperfiltration causes a strain on the kidney filtering units, eventually resulting in a structural change in the kidney, and oxidative stress damages the tissues and cells. The newly damaged kidney, with time, will lose its efficiency, affecting the filtration of the human body, and increasing the ratios of waste in the bloodstream, such as creatinine and urea.

Creatinine is considered a byproduct of muscle metabolism and plays an important part in diagnosing the functionality of the kidney. Its primary usage lies in estimating the glomerular filtration rate, the most important measurement unit for kidney health. Creatinine is formed by the breakdown of creatine and phosphocreatine in skeletal muscles at a constant rate related to muscle mass. The kidney plays a primary role in creatinine clearance, so its levels in the blood are directly related to kidney function and efficiency. Creatinine levels are stable in healthy individuals, marking an equilibrium between production and excretion. On the other hand, when kidney function decreases, creatinine clearance decreases as well, leading to higher levels of creatinine in the blood, indicating an early detection of kidney dysfunction [10].

The article in [11] enlightens one of the significant complications related to diabetes, diabetic nephropathy, which is considered a leading cause of end-stage renal disease. The infection is measured by impaired renal function; the elevated levels of biomarkers such as serum urea, creatinine, fasting blood sugar and serum creatinine are particularly relevant in assessing renal failure in individuals with diabetes. The article proved a relationship between fasting blood sugar levels and renal dysfunction, as elevated fasting blood sugar levels correlate with increasing kidney damage, which is observed by the increased serum urea and creatinine. The study showed a weak positive correlation, $r = 0.28$, between fasting blood sugar and serum creatinine, where r ranges from one to minus one. Pointing out that poorly controlled fasting blood sugar will cause a decrease in the functionality of the kidneys. Also, high blood sugar levels lead to chronic hyperglycemia, resulting in pathological changes in the kidneys, such as glomerular hyperfiltration and eventual nephron damage. The study found higher mean creatinine levels for the diabetic patients (1.13 ± 0.77 mg/dL) compared to the non-diabetic patients (0.89 ± 0.21 mg/dL); the numbers prove the contribution of diabetes to kidney dysfunction. Also, the study found that male diabetics tend to have higher serum creatinine levels than females, likely due to greater muscle mass and associated creatinine production in males.

III. METHODOLOGY

The research methodology is divided into four main stages, as Fig. 1 indicates. Starting with the data collection, which involved collecting a dataset from a private medical center in Palestine. In the second stage, the data cleaning and preparation, the null, invalid, or empty rows were processed either by deleting or replacing with the mean values to ensure data integrity. In the third stage, the exploratory data analysis, multiple visualization techniques were applied to present the outcomes from the dataset using Python software. In the final stage, the modeling stage involved using two different machine learning models to compare the accuracy of each concerning the data cleaning process.

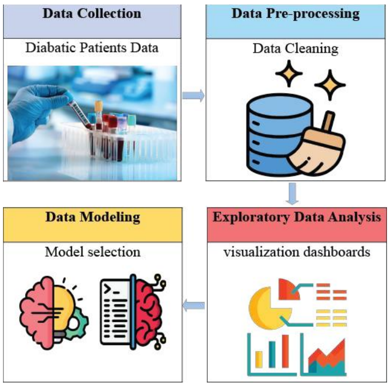

Figure 1- The research methodology used in processing diabetes dataset.

A. Data Collection

The dataset was collected from a private Palestinian medical center. The raw dataset contains 499 records of patients. The data cleaning resulted in two new data frames based on the cleaning method, where the first data frame was reduced to 310 records, divided into 142 records for males and 168 records for females; the second data frame was reduced to 496 records, divided into 241 records for males and 255 records for females. The main features of the dataset are: 1) patient ID, 2) Sex, 3) Marital status 4) Creatinine, 5) Creatinine status, 6) Fasting Blood Sugar, 7) Fasting Blood Sugar status, 8) Hemoglobin A1C, 9) Hemoglobin A1C status 10) High-density lipoproteins (HDL), 11) High-density lipoproteins status, 12) Low-density lipoproteins (LDL), 13) Low-density lipoproteins status, and the size of the Excel file was 22.9 KB.

B. Data Pre-processing

The cleaning and preparation were implemented by the use of Python programming software using multiple libraries as follows: NumPy to handle mathematical operations and arrays, Pandas for data frame manipulation, and Seaborn for a high level of visualization. The data cleaning started by first uploading and reading the data file, then renaming the dataset with a new naming system to make the data more readable and understandable. After loading the file, two methods for cleaning were applied as mentioned. Method: one omitted the rows containing invalid, missing, and null values. Alternatively, method two replaced the missing and null values with the mean value of each feature after omitting the invalid data; the dataset contained negative values, inconsistencies, text, and emojis. After the cleaning, the anomalies and outliers were handled in both frames by applying the Z-score method, where any value, despite its vector value, larger than the standard deviation by three is considered an outlier [12].

C. Exploratory Data Analysis

The visualization of data was performed using Python programming software by implementing various types of visualization expressions to present the dataset findings, such as scatter plots and bar plots. A scatter plot was used to demonstrate the distribution of the features based on gender, whereas the bar plot was used to compare the variation in infection rates based on gender.

D. Data Modeling

The original data frame contains thirteen columns presenting information about diabetic individuals, such as ID, gender, test levels, and normality flags. However, none of these columns directly address the chronic kidney infection presence or status, which is the targeted value. To overcome such a lack of data, the need to create a new column representing the status of chronic kidney infection is urgent. The new feature will be mapped based on the following relations, starting with creatinine. Creatinine levels are direct markers of kidney function, where higher levels can indicate reduced kidney filtration rates [10]. Fasting blood sugar and hemoglobin A1c, where poor glycemic control is strongly associated with diabetes, a leading cause of chronic kidney disease [9][13]. Ending with high-density lipoprotein (HDL) and low-density lipoprotein (LDL) cholesterol, where the HDL and LDL levels are associated with cardiovascular diseases and kidney damage. Which results from oxidative stress and inflammation [14]. HDL is considered to be protective due to its anti-inflammatory and antioxidative properties; alternatively, LDL contributes to oxidative stress, inflammation, and atherosclerosis, increasing the risk of both cardiovascular disease and kidney damage [15]. After the new mapping, new data frames are generated; the first one is sized 310x14, and the second is 496x14. In this paper, two primary machine learning algorithms, namely, k-nearest neighbors and random forest, were developed for the early prediction of kidney dysfunction in diabetic patients based on threshold-derived labels using fasting blood sugar and creatinine levels. Additionally, logistic regression and support vector machine were utilized for comparative evaluation.

IV. EXPLORATORY DATA ANALYSIS AND VISUALIZATION RESULTS

Various types of visualizations were applied to the diabetes dataset for both genders, providing a clear understanding of the results. The results section in this paper has been divided into two main parts: female results after cleaning and male results after cleaning, as follows.

A. Female results after cleaning

The visualization in Fig. 2 shows the medical test results. For female patients, with three colors, each color reflects the level of the medical test. The green color is used for the normal range (N), the red color represents a high range (H), and the yellow color indicates a low range (L)

Fig. 2 (e) and (f) represent the Hemoglobin A1C test levels for female patients. Figure 2(e) presents results obtained after data cleaning using Method One, while Figure 2 (f) shows results processed with Method Two. A noticeable difference between the figures is presented, where Figure 2(e) shows a lower density of data than Figure 2 (f). In both the figures, a high frequency of high range levels is observed; in the first data frame (310×14), 90 female patients have high high-density lipoprotein levels; also, in the second data frame (496×14), 122 female patients have high high-density lipoprotein levels as well.

Fig. 2 (g) and (h) represent the low-density lipoprotein test levels for female patients. Fig. 2(g) presents results obtained after data cleaning using Method One, while Fig. 2(h) shows results processed with Method Two. No data points exceed the 100 mg/dL low threshold in both figures, indicating that all patients fall in the low LDL range. On the other hand, a noticeable difference between the figures is presented, where Figure 2(g) shows a lower density of data than Figure 2(h). Fig. 2 (i) and (j) represent the high-density lipoprotein test levels for female patients. Figure 2(i) presents results obtained after data cleaning using Method One, while Fig. 2 (j) shows results processed with Method Two. No data points exceed the 80 mg/dL high threshold in both Fig. 2(i), indicating that all patients fall in the high HDL range. On the other hand, one point exceeded the high HDL range in Figure 2 (j). A noticeable difference between the figures is presented, where Figure 2(i) shows a lower density of data than Figure 2 (J).

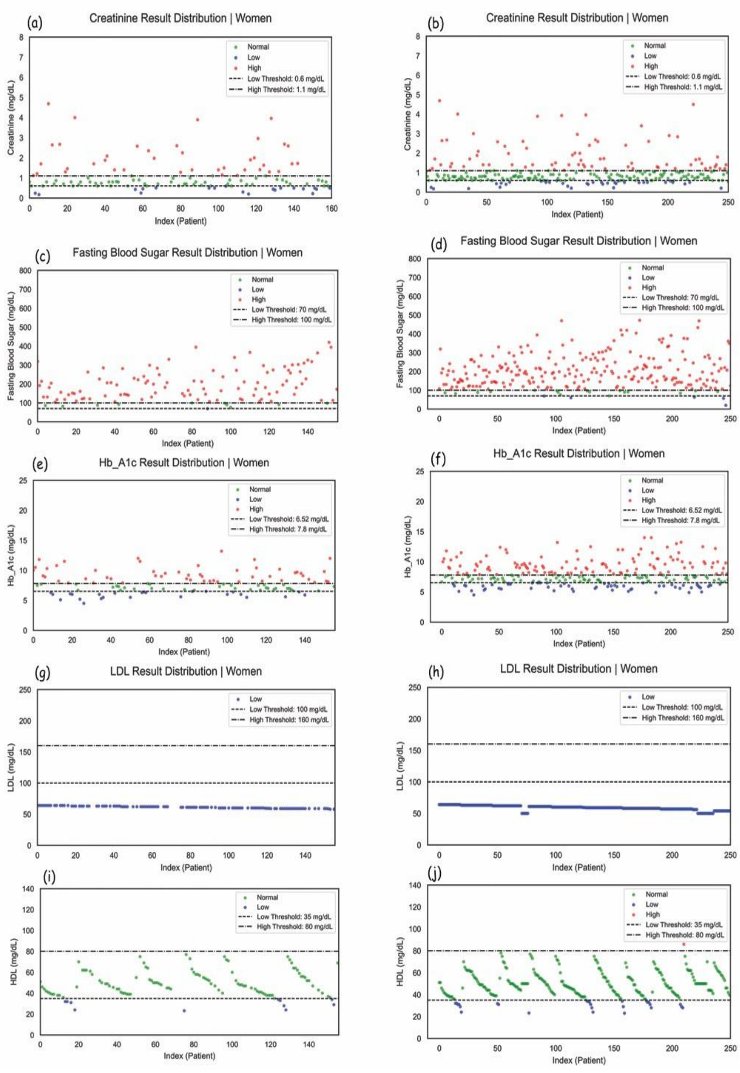

Figure 2- Medical tests visualizations of female patient show creatinine (a), fasting blood sugar (c), hemoglobin A1c (e), LDL levels (g), and HDL levels (i).

B. Male results after cleaning

The visualization in Fig. 2 shows the medical test results. For male patients, with three colors, each color reflects the level of the medical test. The green color is used for the normal range (N), the red color represents a high range (H), and the yellow color indicates a low range (L).

Fig. 3 (a) and (b) represent the creatinine test levels for male patients. Figure 3(a) presents results obtained after data cleaning using Method One, while Figure 3(b) shows results processed with Method Two. A noticeable difference between the figures is presented, where Figure 3(a) shows a lower density of data than Figure 3(b). In both the figures, a high frequency of high range levels is observed, where in the first data frame (310×14), 28 male patients have high creatinine levels; also, in the second data frame (496×14), 44 male patients have high creatinine levels as well. Compared to male patient records, female patients in this dataset exhibited higher creatinine values, exceeding the clinical threshold, more frequently than male patients.

Fig. 3 (c) and (d) represent the fasting blood sugar test levels for male patients. Figure 3(c) presents results obtained after data cleaning using Method One, while Figure 3(d) shows results processed with Method Two. A noticeable difference between the figures is presented, where Figure 3(c) shows a lower density of data than Figure 3(d). In Fig. 3(d), a higher frequency of high-range levels is observed, with some extremely high values exceeding 450 mg/dL. In the first data frame (310×14), 122 male patients have high fasting blood sugar levels; also, in the second data frame (496×14), 203 male patients have high fasting blood sugar levels as well. Compared with male patient records, females tend to have slightly higher fasting blood sugar.

Fig. 3 (c) and (d) represent the fasting blood sugar test levels for male patients. Figure 3(c) presents results obtained after data cleaning using Method One, while Figure 3(d) shows results processed with Method Two. A noticeable difference between the figures is presented, where Figure 3(c) shows a lower density of data than Figure 3(d). In Fig. 3(d), a higher frequency of high-range levels is observed, with some extremely high values exceeding 450 mg/dL. In the first data frame (310×14), 122 male patients have high fasting blood sugar levels; also, in the second data frame (496×14), 203 male patients have high fasting blood sugar levels as well. Compared with male patient records, females tend to have slightly higher fasting blood sugar.

Fig. 3 (e) and (f) represent the Hemoglobin A1C test levels for male patients. Figure 3(e) presents results obtained after data cleaning using Method One, while Figure 3 (f) shows results processed with Method Two. A noticeable difference between the figures is presented, where Figure 2 (e) shows a lower density of data than Figure 2 (f). In both the figures, a high frequency of high range levels is observed; in the first data frame (310×14), 75 male patients have high hemoglobin A1C levels; also, in the second data frame (496×14), 124 male patients have high hemoglobin A1C levels as well. Compared with male patient records, females tend to have higher Hemoglobin A1C test levels in the first data frame and slightly equivalent numbers in the second data frame.

Fig. 3 (g) and (h) represent the low-density lipoprotein test levels for male patients. Fig. 3 (g) presents results obtained after data cleaning using Method One, while Fig. 3 (h) shows results processed with Method Two. No data points exceed the 100 mg/dL low threshold in both figures, indicating that almost all patients fall in the low LDL range, except two records, to are slightly higher than the high LDL range in the second data frame (496x14). On the other hand, a noticeable difference between the figures is presented, where Figure 3 (g) shows a lower density of data than Figure 3 (h). Compared with female patient records, both genders have, in general, a low LDL range. Fig. 3 (i) and (j) represent the high-density lipoprotein test levels for male patients. Figure 3 (i) presents results obtained after data cleaning using Method One, while Fig. 3 (j) shows results processed with Method Two. A noticeable difference between the figures is presented, where Figure 3(i) shows a lower density of data than Figure 3 (J). In both the figures, a frequency of high range levels is observed; in the first data frame (310×14), 24 male patients have high-density lipoprotein levels; also, in the second data frame (496×14), 38 male patients have high high-density lipoprotein levels as well. Compared with female patient records, females tend to have lower high-density lipoprotein levels in the first data frame and none in the second data frame. Based on this, we conclude that the males tend to have higher high-density lipoprotein levels than female patients.

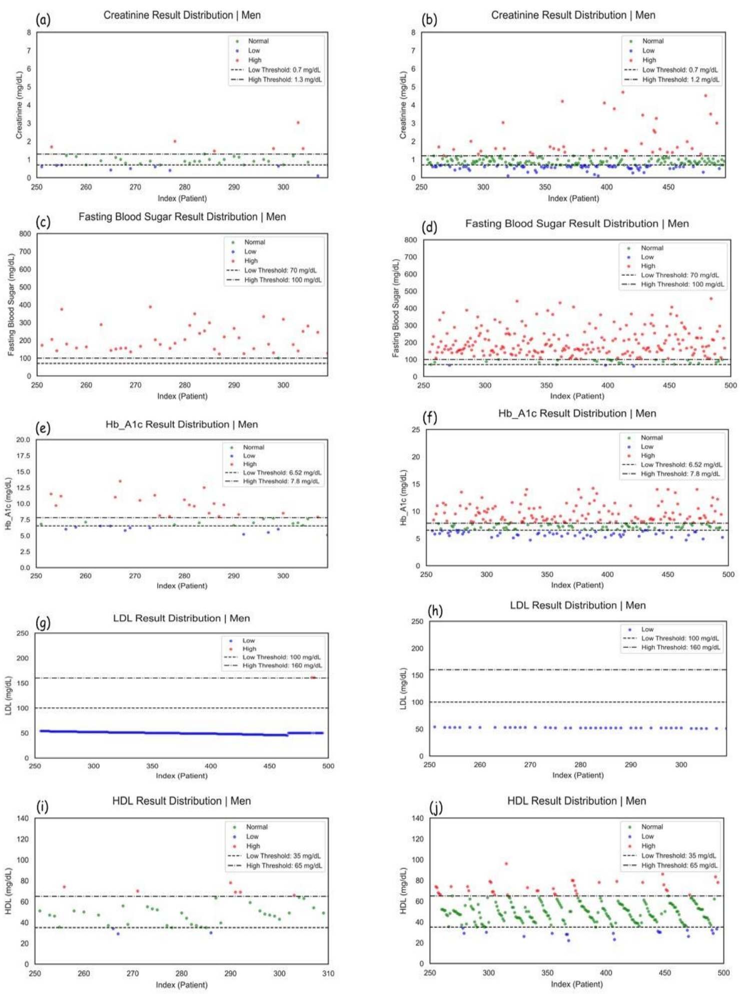

Figure 3- Medical tests visualizations of male patient show creatinine (a), fasting blood sugar (c), hemoglobin A1c (e), LDL levels (g), and HDL levels (i).

C. Modeling results

Fig. 4 (a) and (b) represent the distribution of chronic kidney infection by gender. Figure 4 (a) presents results obtained after data cleaning using Method One, while Fig. 4 (b) shows results processed with Method Two. A noticeable difference between the figures is found, where Figure 4 (a) shows a lower size of data compared with Figure 4 (b). In both figures, males have a higher number of non-infected cases, while females have a higher number of infected cases. Pointing to higher infection ratios between females compared to males. Despite the inequality in size between the two data frames, the data remains consistent. Concluding that the influencing factors are stable across different datasets or time periods.

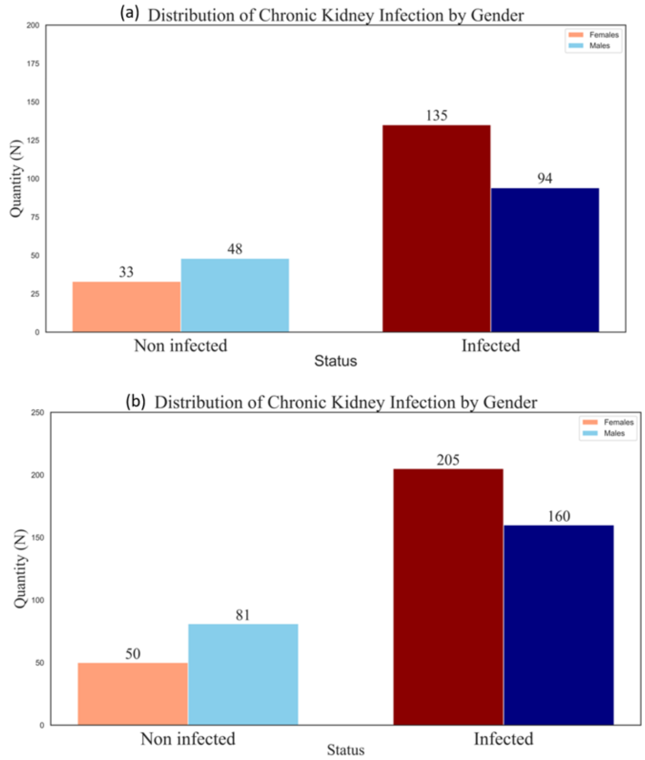

Figure 4- Distribution of Chronic Kidney Infection by Gender (a), data frame one (310x14) (b), data frame two (496x14).

The study focused on the early prediction of kidney dysfunction in diabetes patients based on fasting blood sugar and creatinine status. To achieve such a target, machine learning models were applied. The used models were the K-Nearest Neighbors Algorithm, the Random Forest Algorithm, Logistic Regression, and Support Vector Machine, scoring on the first data frame (310x14) RF achieved the highest performance with a scoring accuracy of 98.39% and a perfect ROC-AUC of 1.0, while KNN scored only 85.48% accuracy with a ROC-AUC of 0.86. On the second dataset, RF accuracy remained unchanged with a value of 98.39% and ROC-AUC = 1.0; on the other hand, the KNN accuracy improved to 93.0% with ROC-AUC = 0.93 due to balanced gender distribution. LR and SVM also achieved competitive results. On the first dataset, LR obtained 95.16% accuracy with a ROC-AUC value of 0.977, while SVM accuracy remained unchanged with a value of 91.93% and ROC-AUC = 0.963. On the second dataset, LR again achieved 95.16% and ROC-AUC = 0.977, while SVM recorded 91.93% accuracy and ROC-AUC = 0.963.

V. CONCLUSIONS

The results of this study highlight the effects of data-cleaning methods on the consistency and reliability of medical test analysis. Two methods were applied: method one and method two. Method one omitted missing and invalid entries, resulting in inconsistencies in the dataset, making the judgment on the results more challenging, as some trends appeared less clear or biased by gender, as male and female records dropped from (241, 255) to (168, 142). On the other hand, method two replaced missing and invalid entries using the mean value, resulting in non-biased trends and a more consistent dataset. Also, machine learning algorithms such as K-Nearest Neighbors and Random Forest were applied to predict early kidney dysfunction in diabetes patients. The K-Nearest Neighbors and the Random Forest algorithms achieved high accuracy results, with Random Forest holding the highest accuracy score. However, the accuracy varied based on data cleaning methods. The study also pointed to the distribution of kidney dysfunction indicators among female patients being higher than among male patients. Where high creatinine, fasting blood sugar, and hemoglobin A1C levels.

In conclusion, this study enlightened the importance of data preprocessing methods in medical research. The omission of data in Method One biased the analysis, while the mean estimation in Method Two provided a more stable, consistent, and reliable dataset. The output of this finding highlights the necessity of careful data handling. Given the higher risk of kidney dysfunction among female diabetes patients, early detection and intervention strategies must be prioritized.

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